The Reheat Concept: The Proven Pathway to Ultralow Emissions and High Efficiency and Flexibility

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Reheat combustion has been proven now in over 80 units to be a robust and highly flexible gas turbine concept for power generation. This paper covers three key topics to explain the intrinsic advantage of reheat combustion to achieve ultralow emission levels. First, the fundamental kinetic and thermodynamic emission advantage of reheat combustion is discussed, analyzing in detail the emission levels of the first and second combustor stages, optimal firing temperatures for minimal emission levels, as well as benchmarking against single-stage combustion concepts. Second, the generic operational and fuel flexibility of the reheat system is emphasized, which is based on the presence of two fundamentally different flame stabilization mechanisms, namely, flame propagation in the first combustor stage and autoignition in the second combustor stage. This is shown using simple reasoning on generic kinetic models. Finally, the present fleet status is reported by highlighting the latest combustor hardware upgrade and its emission performance.
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1 Introduction

In the mid 1990s, Alstom introduced two similar sequential combustion gas turbines: the GT24 for the 60 Hz market and the GT26 for the 50 Hz market. Since its first launching in 1995 [1], the advanced class GT24/GT26 engines have demonstrated that this technology platform does offer superior operating flexibility, low emissions, and high part-load efficiency with world-class levels of reliability and availability.

The main technology differentiator of Alstom's GT24/GT26 (Fig. 1) gas turbines is the sequential combustion principle, which was already introduced in 1948 into the market as a way of increasing efficiency at low turbine inlet temperature levels. The GT24/GT26 combustion system is based on a well-proven Alstom combustion concept using the environmental (EV) burner in an annular combustor followed by the SEV (sequential environmental) burner in the second combustion stage (see Fig. 1). This dry low NO_x EV burner has a long operating history and is used in the whole Alstom gas turbine range. Sequential combustion, "the reheat principle for gas turbines," had already been applied to earlier (Brown Boveri) engines but using two side-mounted silo combustors. Integrating the concept of dry low NO_x EV burner and sequential combustion into a one shaft engine resulted in the GT24/GT26—a machine with a high power density and a small footprint. With a pressure ratio of $>30:1$, the compressor delivers nearly double the pressure ratio of a conventional compressor.

The compressed air is heated in a first combustion chamber (EV combustor) by adding about 50% of the total fuel (at base load). After this, the combustion gas expands through the high-pressure (HP) turbine, which lowers the pressure by approximately a factor of 2. The remaining fuel is added together with some additional cooling air in the second combustion chamber (SEV combustor), where the combustion gas is heated a second time to the maximum turbine inlet temperature and finally expanded in the four-stage low-pressure (LP) turbine.

Relative to a conventional nonreheat cycle, the same specific power output is achieved at lower turbine inlet temperature. This is illustrated in Fig. 2.

2 Basic Reheat Features for Low Emissions

The low emission levels, which can be achieved with a reheat system, are the combined effect of three key mechanisms: First, a reheat combustor makes a more efficient use of the oxygen by burning twice in the lean premix mode. Second, there exists a chemical advantage of reheat combustion, which can be exploited. Third, the flame stabilization by autoignition leads to increased flexibility, which allows operating in low emission mode at a wide load range by avoiding high peak flame temperatures T_{flame}, which lead to exponential increase in NO_x, for both combustors. We explain these effects in detail below.

2.1 Impact of Lower O_2 Level. The reheat concept also makes very efficient use of the O_2 from the air by operating the second stage at approximately 15% O_2 at the inlet. Overall this leads to very high power densities. The O_2 content in the exhaust is in the order of ~10%, yielding a beneficial normalization factor (for 15% O_2) of $(21-15\%)/(21-10\%\text{-})$ ~0.55 for the absolute NO_x, as shown in Fig. 3 on the right hand axis. The correction factor decreases with higher temperature (~ consumed fuel ~O_2) and is lower for the reheat engines due to its increased O_2 consumption.

This leads to varying absolute levels for a fixed "15%" guarantee value depending on reheat or nonreheat combustion and T_{flame} and allows higher absolute emission values after two combustion chambers. This correction has been introduced to make a fair comparison of power plants based on their emissions relative to consumed air or fuel. Emissions quantified in the widely used

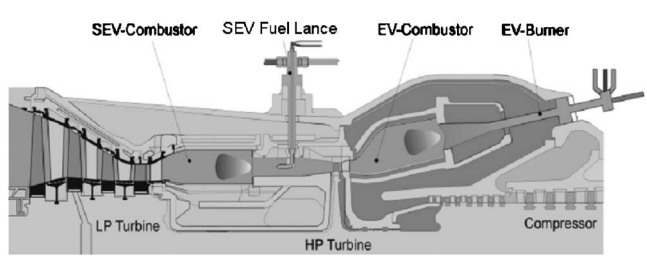

Fig. 1 GT24/GT26 sequential combustion system

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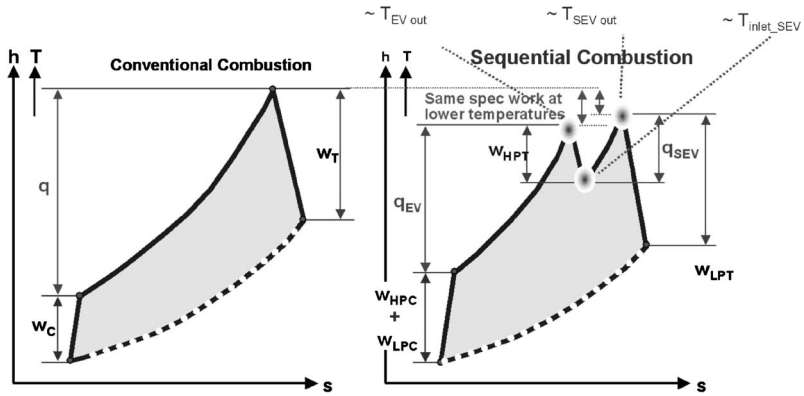

Fig. 2 Principle of the reheat cycle compared to a standard GT cycle

emission index defined as (atmospherically produced) $g(NO_2)/kg$ fuel are reduced accordingly.

2.2 NO_x Characteristics of Reheat Combustion. A second reason for the low NO_x emission is the combustion in the SEV at reduced O_2 levels and increased H_2O contents from the first combustor ($\sim 5\%$). This allows the SEV combustor to operate at higher firing temperatures and produce less NO_x than an EV combustor would produce at the same temperature.

Results of CHEMKIN [2] simulations using the GRI-MECH [3] for the different flame types are shown in Fig. 3, where the laminar flame speed module PREMIX has been used for the EV [4] and the plug flow reactor (PFR) module for SEV. A detailed network model based on PREMIX and PFR has been developed for an accurate emission prediction for realistic engine conditions, but its details are out of the scope of this paper.

A simple CHEMKIN calculation shows that NO_x produced in the second combustor is essentially unaffected by NO_x levels arising from the EV combustor as long as they are of moderate magnitude. The SEV calculation seems simply to be offset by the amount of EV NO_x given at the start. Therefore, a chemical model of the engine treats both combustors as independent NO_x producers. Although the GRI is assumed to describe the SEV combustion sufficiently, further efforts are taken to investigate possible effects of NO_x and water as well as reduced O_2 content in ongoing projects in connection with shock tube studies.

Shown in Fig. 3 are the NO_x contributions versus T_{flame} of a single combustor (EV) at approximately 30 bars and of a SEV-type combustor at approximately 15 bars, as well as a combination of the two with fixed T_{flame} for the first combustor (EV) for the idealized situation of perfect mixing without heat losses. NO_x emissions are given on an absolute scale normalized to the value for a single combustor (EV) at T_{ref}. This allows the SEV contribution (at reduced O_2) to be shown on the same plot. The curves are analytical fits to CHEMKIN II results of the GRI3.0. The reheat engine results in lower emissions at the same T_{flame}. The calibration factor for 15% O_2 refers to the right hand scale. The NO_x emissions of the full reheat engine are taken as the sum of the absolute emissions of the two combustors assuming no interaction of NO_x from the first combustor on the second.

The reduced NO_x production of the reheat combustor is among other possible factors due to the decreased concentration of O radicals, which is influenced by the reduced O_2 and the increased

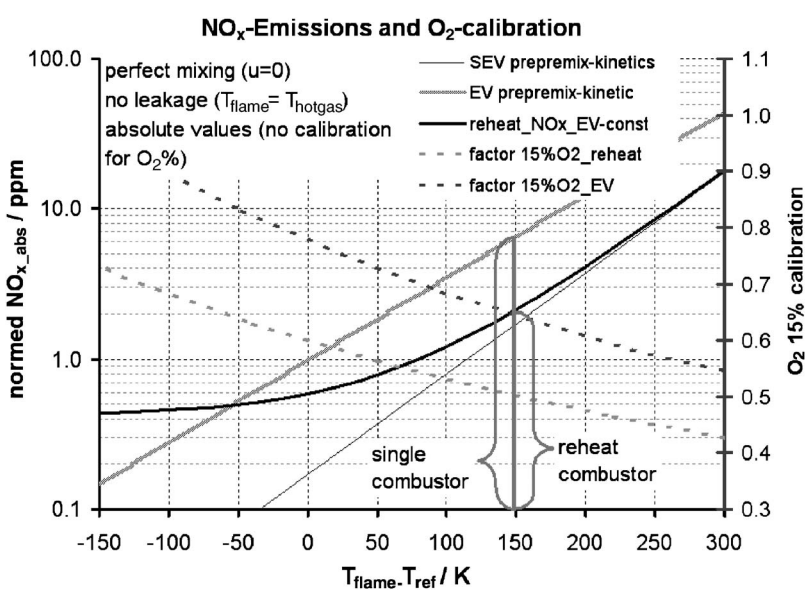

Fig. 3 NO_x emissions on log scale (normalized, noncalibrated) for a single combustor (EV), the SEV combustor, and a reheat combustion system. The factors calibrating to 15% O_2 depend on T_{flame} and are shown in green and refer to the right hand scale.

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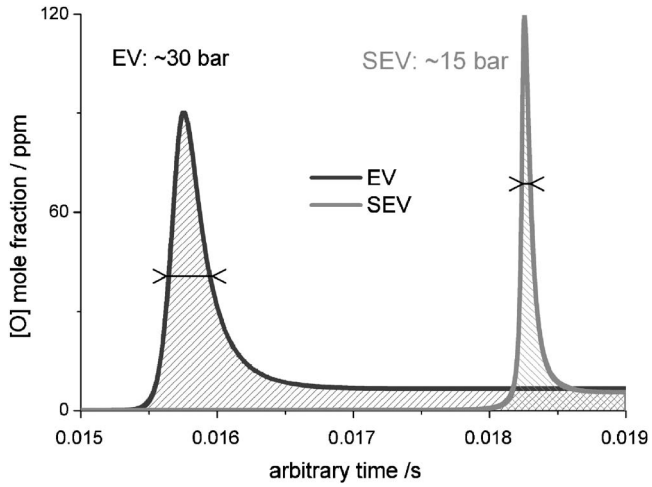

Fig. 4 O-radical concentration profiles for EV and SEV flames (taken from PREMIX and PFR calculations) for similar T_{flame}. The arbitrary time scale extends over 4 ms. The shaded area in the integral $\int t[O]_{dt}$ determines the most important NO_x formation routes and is approximately three times higher in the EV than in the SEV.

H_2O content. The O radicals are an essential part of the main route to produce NO_x by oxidizing the otherwise stable N_2 molecule according to the kinetic rate law:

$$\frac{d[\mathrm{NO}]}{dt} = k(T,p)[\mathrm{N_2}][\mathrm{O}] \tag{1}$$

$$[\mathrm{NO}] = k(T,p)[\mathrm{N_2}]t[\mathrm{O}] \tag{2}$$

Since k and the N_2 concentration do not vary much for given temperatures and pressures, the NO_x formation depends mainly on the time t and the concentration O radicals. The time profiles of this species are plotted in Fig. 4, revealing the integral to be approximately three times smaller, which is the combination of short residence times and low O concentration due to H_2O and O_2 contents of the vitiated air as explained above. For such laminar cases, the prompt NO_x formation due to the O radicals is estimated to be approximately three to six times smaller at given T_{flame} considering O-concentration profiles. To invoke the often invoked reburn chemistry to explain the low NO_x emissions seems not to be necessary and is believed to be unlikely since the heat release is occurring entirely in fuel lean conditions.

Another factor reducing SEV-NO_x emissions is the significantly reduced residence time in the SEV compared to the EV. Some additional benefit might arise from the fact that the fast SEV gas velocities of part of the rich combustion products in the SEV result in faster quenching of this hot pockets and subsequently less time to produce NO_x. It can be summarized that the NO_x production in the SEV starts at a significantly lower value.

2.3 Reheat Combustion With Autoignition. The reheat concept allows a very flexible operation of the engine, enabling us to find an optimized operating range for the given conditions and fuel compositions. The reason for that is described in the following section.

While the EV inlet temperature T_{inlet_EV} is determined by the high-pressure compressor, in the SEV the inlet temperature T_{inlet_SEV} is governed by the outlet temperature of the HP turbine. The SEV flame temperature, T_{flame_SEV}, determines the inlet temperature of the LP turbine. Accordingly, the HP-turbine inlet temperature (and, along with it, T_{inlet_SEV}) is controlled by the EV flame temperature, T_{flame_EV}.

The reactor inlet temperature T_{inlet} in the chemical model is derived from mixing cold fuel and hot combustion air before the reaction, yielding an effective T_{mix} as T_{inlet}. The shorter residence time of the products in the combustor is accounted for in the model.

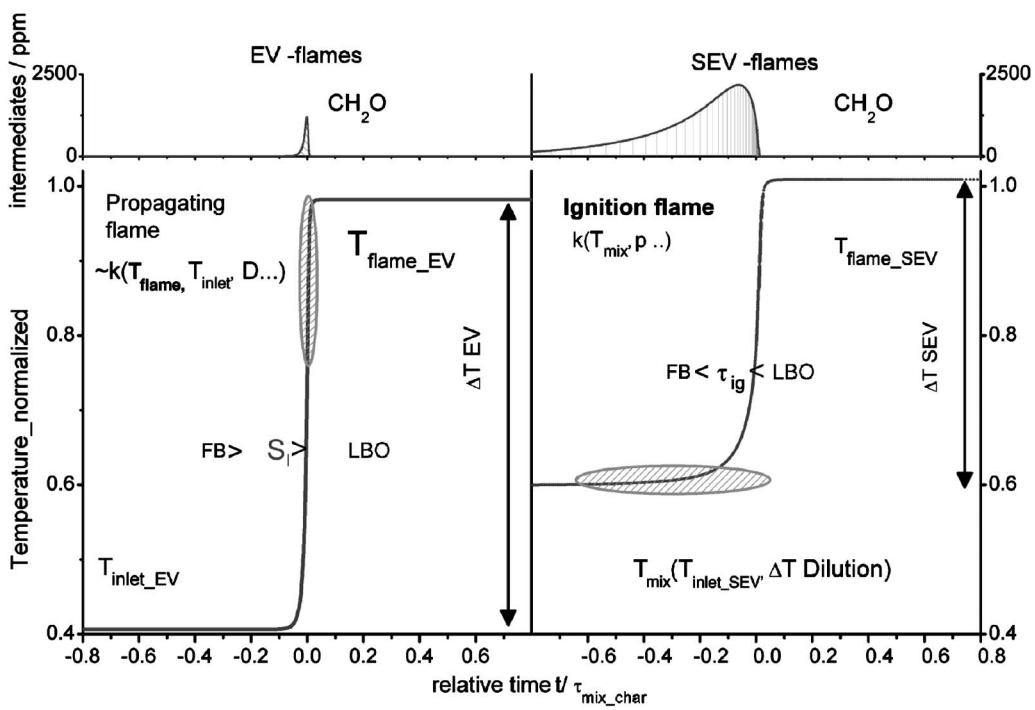

Fig. 5 Schematic diagram explaining the physics and chemistry of the EV flame (left) and the SEV flame (right) for natural gas. The intermediates indicating preflame zones (CH_2O) are displayed in the upper graphs. The shaded areas indicate the most relevant regions of reactivity. The different time axes were given in reverence to the typical mixing times.

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The higher $T_{\text{inlet_SEV}}$ results in a much higher reactivity of the fresh gases, a different mechanism being important for the description of the flame. This higher reactivity is partly compensated by higher burner velocity leading to shorter characteristic mixing times $\tau_{\text{mix_SEV}}$ ($\sim$ ignition time $\tau_{\text{igntion}} \sim 1-2$ ms) available for mixing for the SEV combustor, leaving some residual unmixedness in the flame region compared to the EV flame, where a characteristic mixing time $\tau_{\text{mix_EV}}$ ($\sim 5-10$ ms) is given by the traveling time from fuel injection to flame front.

It should be pointed out that Fig. 3 indicates ideal conditions. For a real system, effects of local unmixedness and heat losses need additional consideration.

This high reactivity at high combustor inlet temperatures in the SEV has the consequence that flame stabilization occurs always under premixed conditions by autoignition, and no piloting is needed to stabilize the SEV flame. The T_{flame} of the two combustors are ideally optimized primarily with respect to performance and turbine lifetime, but also at a given load for equal amounts of NO_x produced, deriving an absolute minimum in engine emissions.

2.4 Optimal Firing of Combustor for Minimal Emissions and Maximum Flexibility. The reheat concept enables us to operate the two combustors at different temperatures with only little impact on the overall power output. This increases the flexibility of the combustion system, allowing us to burn more or less reactive fuels in the same engine by adjusting the relative load between the two combustors and to mitigate ranges where operation would become difficult. This can be utilized for different fuels (C2+ or even H_2 containing fuels), exhaust gas recirculation, or changes in ambient conditions.

In general, the engines are adjusted so that the first combustor runs at low firing temperatures, keeping the emissions as low as possible for conventional premix combustion, while the second combustor is operated at higher temperatures, producing similar emissions (on absolute scale) due to the reasons mentioned above. Note that while the EV runs more or less at constant conditions (see Fig. 10), keeping low NO_x even at very low loads, the loading is achieved by varying the fuel input in the SEV starting at as low as $\sim$25% GT load (as well as increasing the mass flow of air). The high $T_{\text{inlet_SEV}}$ allows operation without piloting for the entire operation range, entirely relying on autoignition for the SEV flame.

The conceptual difference between EV and SEV flames is presented in Fig. 5. The flame has to be stabilized between limits for flashback (FB) and lean blowoff (LBO) determined by the laminar flame speed S_L for the EV flame, which depends on conditions around the adiabatic T_{flame} and less on T_{inlet}. The SEV flame is dominated by the chemistry occurring in the preflame zone at T_{inlet}. This includes the buildup of a radical pool up to a critical threshold. The flame position is here determined by the ignition time. To compare the two different combustion regimes, the time has been scaled to zero for the region of maximum heat release and normalized to the characteristic mixing times as defined before.

For the inlet conditions of the EV, autoignition times are much too long to play a role even for the most reactive fuels. Flame propagation is the leading mechanism for a stationary reaction zone stabilization, as can be calculated with PREMIX for the laminar case (laminar flame speed S_L). The description of the turbulent flame propagation is derived from that concept. As for every conventional premix burner, the EV flame stabilization happens within the boundaries of flame blowoff at a low flame temperature limit and flame FB at a high flame temperature limit. These flame propagation limits are sketched on the left side of Fig. 5. The reactivity is controlled by transport across the flame front and determined by the kinetic rates at high temperatures close to the adiabatic one. Intermediates are abundant only close to the flame front. The reactivity (Fig. 6) depends on the kinetics of the fuel

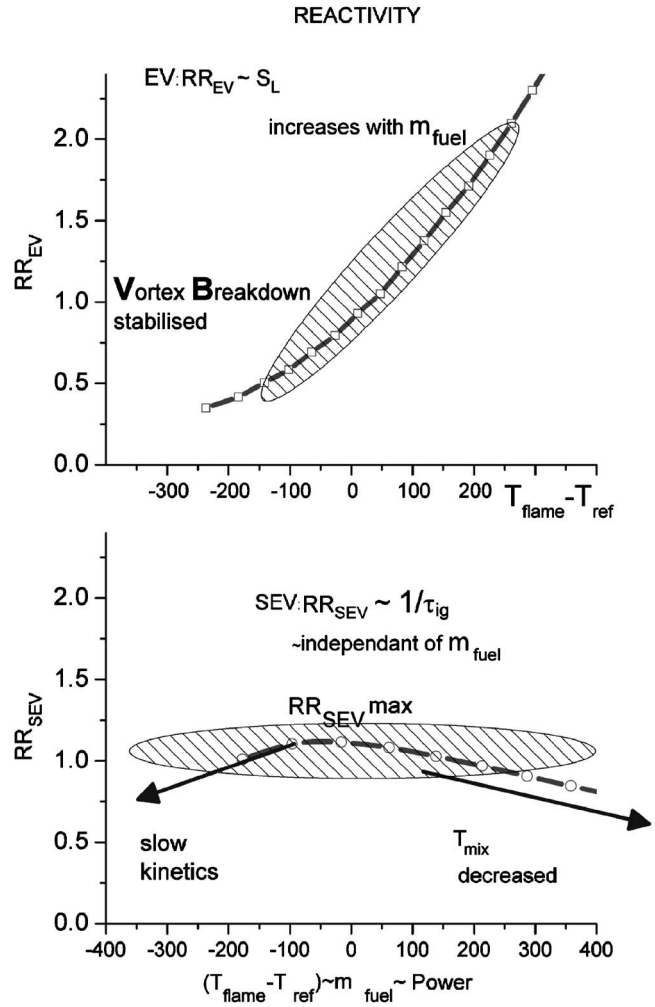

Fig. 6 EV and SEV reactivities as defined by Eqs. (3) and (4) for varied fuel contents and constant T_{inlet} (compare with Fig. 8). The shaded area defines an operational range.

and its transport properties at high temperatures close to the final $T_{\text{flame_EV}}$ and only slightly on pressure and $T_{\text{inlet_EV}}$. For the EV, the relative reactivity RR_{EV} is defined with reference to one operating point as the ratio of laminar flame speeds in

$$\text{RR}_{\text{EV}} = S_L / S_{L-\text{ref}} \tag{3}$$

This flame stabilization is fundamentally different for the SEV flame, which is controlled by autoignition (calculated with PFR) arising from reactive intermediates forming and accumulating up to a critical threshold before the flame front. The chemistry at lower temperatures ($\sim T_{\text{inlet_SEV}}$) is responsible for the reactivity. This means that in the engine the SEV flame is controlled and governed by the $T_{\text{flame_EV}}$, while $T_{\text{flame_SEV}}$ has only little influence on the flame position. The relative reactivity RR_{SEV} is defined in Eq. (4) from the ignition times as

$$\text{RR}_{\text{SEV}} = \tau_{\text{ig_ref}} / \tau_{\text{ig}} \tag{4}$$

These reactivity definitions enable comparing combustor specific definitions on one scale (Figs. 6–8).

Plotting RR over T_{flame} ($\sim 1/\phi \sim \lambda$ for lean conditions) in Fig. 6 leads to the contrary effects: For this laminar approximations, the EV flame becomes more reactive with T_{flame}, but the SEV flame becomes less reactive because the cooling effect of adding more cold fuel to the heated air from the first combustor controls the reactivity while the reactivity (toward ignition) is not increased by a higher concentration of fuel. In the very lean region, an increase

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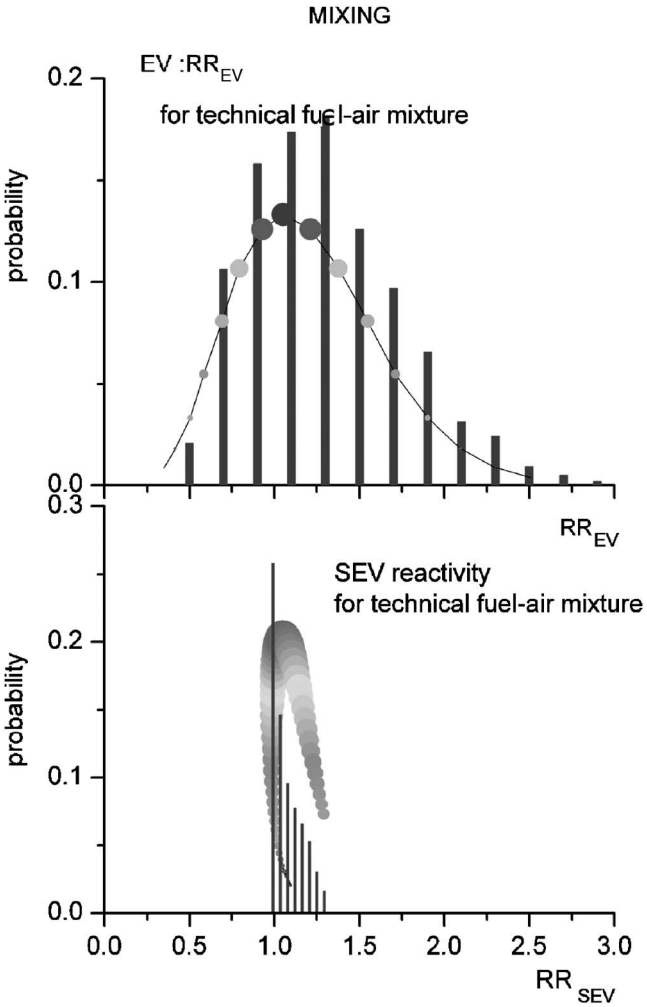

Fig. 7 Probability of reactivity derived from a distribution of fuel for EV and SEV combustors at fixed inlet conditions. The histograms show the probability for a given reactivity, while the bubbles are calculated reactivities for a given fuel-mass ratio.

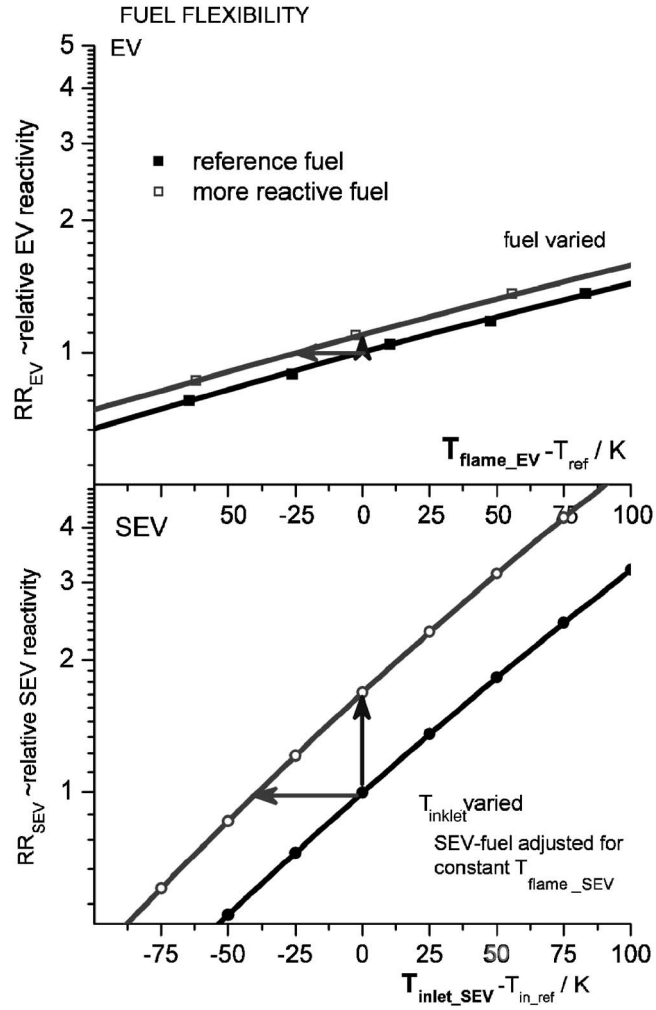

Fig. 8 EV reactivity for varied $T_{\text{flame_EV}}$ (upper graph) for two different fuels on logarithmic scale. In the lower graph, the reactivity of the SEV for varied inlet conditions ($T_{\text{inlet_SEV}}$) is presented, referenced to a standard inlet temperature $T_{\text{in_ref}}$. A reactivity increase due to the fuel is compensated by reduced $T_{\text{inlet_SEV}}$. Note that the RR_{SEV} is plotted versus T_{flameSEV} in Fig. 6.

of fuel molecules can increase RR_{SEV} slightly. Taken together, these effects counterbalance to give a reactivity maximum and, overall, very small dependence on T_{flame}. The small variation for the SEV reactivity with fuel is the cause for the high flexibility of the reheat concept. The reactivity increases with fuel only for very lean conditions. The chemical acceleration of rich mixtures levels off at intermediate levels, and the reduction of T_{mix} (due to the mixing with cold fuel) outruns this effect, slowing down the reactivity toward rich mixtures.

The EV reactivity RR_{EV} varies much more with fuel; therefore, the EV flame position has to be controlled aerodynamically by attaching it to a flame anchoring point, in this case a stagnation point in the flow field (vortex breakdown) of the swirled EV burner. Effects of autoignition in the EV combustor or flame propagation in the SEV are assumed to be minor and are neglected here.

The operating range of a burner is limited by a minimum reactivity ($\sim S_L$ for the EV), causing a LBO of the flame and a maximum acceptable reactivity preventing the flame to flash back. The EV is governed by the aerodynamic stabilization, allowing this combustor to hold the flame at the same positions over a wide range of reactivity.

The reheat combustion system has the particular property that the flame stabilization in both premix combustors, EV and SEV, is controlled by a single physical quantity—the flame temperature of the first EV combustor. This way, the high flexibility of the engine can be maintained by being able to operate the EV over a range of T_{flame}, therefore enabling us to find an optimum HP-turbine outlet temperature for SEV operation. This optimization can be used for the lowest emissions as well as to handle a variety of fuels and inlet conditions.

The SEV flame position is mainly controlled by its chemical reactivity, which is mainly influenced by $T_{\text{inlet_SEV}}$ and the fuel composition. The different control mechanisms of flame positions are utilized in the ALSTOM gas turbine design by attaching the EV flame aerodynamically and by controlling the SEV flame through $T_{\text{inlet_SEV}}$. The fact that the SEV flame is relatively insensitive to T_{flame} leads to a high operational range: SEV load can be increased without increasing FB risk (emissions and pulsations still have to be considered, of course).

This leads to an interesting effect for not perfectly mixed flames. The reactivity probabilities resulting from a given fuel-air unmixedness are plotted in Fig. 7. The same variance in the upper and lower graph lead to a wider distribution of flame speeds for the EV and a very narrow distribution for the SEV (at given HP-turbine outlet temperature). Here the different mechanisms become clear once more by spreading out over a wide range of

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reactivities for the EV flame but only a very narrow range for the SEV-flame.

3 Generic Operational and Fuel Flexibility

3.1 Fuel Flexibility for Highly Reactive Fuels. While the reactivity for the EV is controlled by $T_{\text{flame_EV}}$, the SEV reactivity is mostly controlled by the inlet conditions and $T_{\text{inlet_SEV}}$. This has some interesting conclusions for the engine operation.

1. The SEV can be varied in $T_{\text{flame_SEV}}$ without changing its flame position much.
2. An increased reactivity due to a more reactive fuel can be easily compensated by reducing the $T_{\text{flame_EV}}$ and, therefore, $T_{\text{inlet_SEV}}$.

This is also illustrated in Fig. 8, which demonstrates the fuel flexibility achievable by varying $T_{\text{flame_EV}}$. The increase in reactivity (vertical arrow) due to a more reactive fuel can be compensated by reducing $T_{\text{flame_EV}}$ (horizontal arrow). The reduction of the EV fuel leads to a $T_{\text{inlet_SEV}}$ reduction, which simultaneously reduces the SEV reactivity. In the SEV, a reactivity increase due to the fuel is compensated by reduced $T_{\text{inlet_SEV}}$.

The described insensitivity of flame reactivity to $T_{\text{flame_SEV}}$ allows the inlet temperature of the LP turbine to be constant (in terms of protection for the combustor). Therefore, only little loss in overall power and efficiency due to derating the EV since the power output is largely recovered by adding more fuel to the SEV. This is in contrast to the nonreheat engines, where a derating due to increased fuel reactivity leads to decreased turbine inlet temperature resulting in severe loss of overall power and efficiency.

This key feature can be used to operate the engine [5] such that changes in fuel composition do not impact emission performance and reliability significantly. For example, if for given T_{flame} higher hydrocarbons ("C2+") or even H_2 or CO are added to the fuel, reactivity will increase (i.e., flame speed will increase and ignition time will decrease) and, as a result, the EV combustor will likely produce higher NO_x. Likewise, the SEV combustor may encounter FB risk due to shorter ignition times if no adjustment is foreseen.

For highly reactive fuels, the fuel input in the EV combustor is reduced, which leads to lower SEV burner inlet temperature, to stay below the burner protection limit. This is feasible because the EV burner lean extinction limit is also reduced for increased reactivity in the fuel.

However, a simple adjustment by decreasing the fuel input in the EV combustor and simultaneously increasing the fuel input in the SEV combustor (to keep the total fuel flow constant) will stabilize combustion performance: EV emissions remain low, and SEV flame stabilization is back at its design point. Due to the inherently low emissions of the SEV combustor, the emission penalty will be negligible if the engine is operated in such a way as schematically shown in Fig. 9.

3.2 Dual Fuel Capability. The primary fuel of the GT24/GT26 fleet is gaseous fuel, while fuel oil is often needed as a backup. The sequential combustion system can also be operated with fuel oil, engines in this case being delivered with dual fuel EV and SEV lances instead of the normal gas-only lances. As is normal practice, demineralized water is used for NO_x control, this being combined with the fuel oil in a mixing block before being fed to the lance as an emulsion. The GT24/GT26 engines also have the ability to switch between the two fuels while remaining synchronized to the grid and so providing the operator with arbitration capabilities and full backup security.

3.3 Part-Load Flexibility. The sequential combustion system offers the intrinsic advantage that premix combustion can be applied across the entire load range because the EV combustor is operated at a constant flame temperature through the entire load range, while the (premix) SEV combustor is used to vary loads from approximately 12–100%. This is schematically shown in Fig. 10.

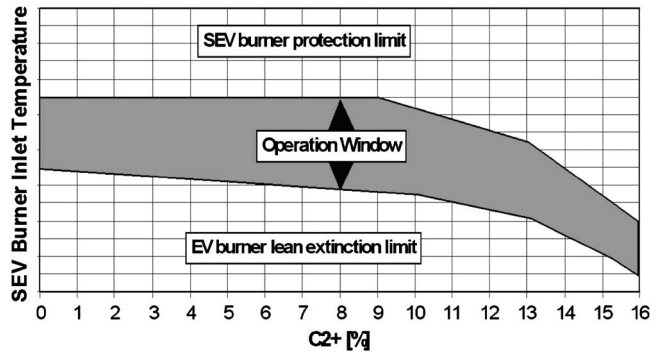

Fig. 9 GT26/GT24 operation concept as a function of fuel reactivity, here given by the content of higher hydrocarbon content (C2+) in the fuel

The so-called "low load operation concept" utilizes the possibility of shutting down the sequential SEV combustor at low part loads and is therefore a unique feature of the GT24/GT26 technology. This concept allows the plant to be operated in a combined cycle mode at a very low combined cycle load (about 15–25%) with the EV combustor operating in the lean premix mode, giving low emission levels as well as a homogenous HP-turbine inlet temperature distribution while keeping the water-steam cycle up and running. The result is a concept that has a number of advantages to the operator because it avoids start-stop cycles and the related cyclic thermal stress to the top and bottom cycle equipment. It enables the operation at low emission levels close to base load levels and reduces cumulative emissions compared to parking a plant at a higher partial load. Also, it assures a homogenous turbine inlet temperature distribution, which is more difficult to achieve for nonreheat engines due to burner piloting at part load.

4 Engine Experience

4.1 Combustor Hardware Status. The continuous drive [6] toward higher performance and lower emissions motivated an upgrade of the EV combustor. The combustor upgrade development, based on the so-called "staged EV burner" is described in detail in a recent paper [7]. Its main feature is the replacement of the central pilot injector by a premix injector. This leads to a burner system with two fuel stages, which can be operated both in the premix mode to vary the fuel profile. In the premix mode, the

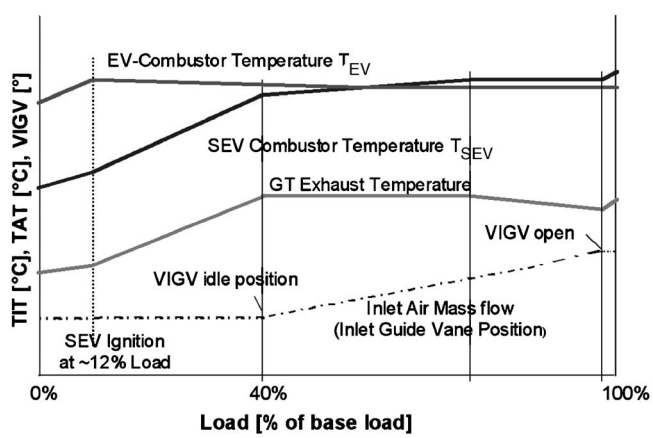

Fig. 10 Schematic of the operating concept of the GT24/GT26 with load variation done through the SEV combustor fuel flow

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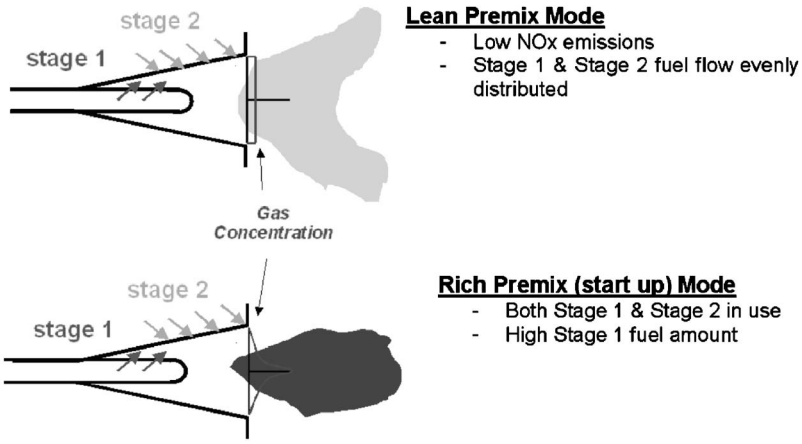

Fig. 11 Staged EV-burner principle. Details of its development are described in Ref. [7].

profile is optimized for minimal emission performance, while in the startup mode, the staging is optimized for maximal stability (see Fig. 11). As a result, this burner has much reduced emission levels in part-load operation, as well as in the base load where the fuel staging between the two stages can always be chosen to be optimal to the base load adjustment point of the engine.

4.2 Emission Characteristics. The emission performance of the staged EV combustor against the standard EV combustor is shown in Fig. 12. This graph shows two effects: First, emission levels are reduced and thus confirm the expected improvement of the staged EV burner. Second, the emission gradient is changed over a wide load range (in this plot between the measured points between 65% and 100% loads). This change in gradient is interesting because it is a direct consequence of the optimization of the EV combustor.

The emission levels of the EV combustor are now so low that the total emissions of the engine start to become dominated by the SEV combustor. As was shown in Fig. 10, the fuel input in the SEV is increased with load and, as a result, now visible in the engine emission performance even if absolute values are very low. Indeed, a more detailed analysis of the combustion system, which includes effects of heat losses and burner imperfections, shows that the relative emission contribution of the SEV combustor increases.

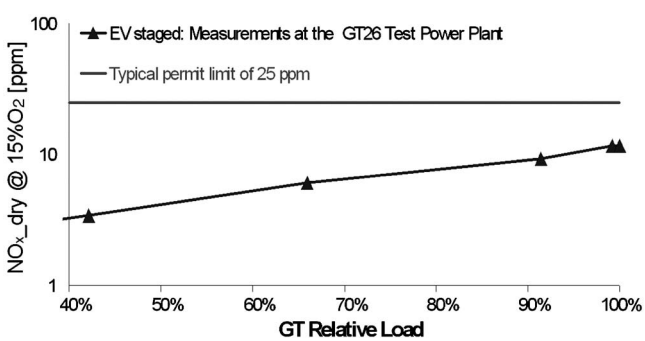

Fig. 12 Emission levels achieved with the staged EV burner on logarithmic scale

5 Summary

In this paper, the key advantages of the reheat system as implemented in the GT24/GT26 engines have been highlighted, namely, its inherently low emission levels as well as its operational flexibility. The contribution of the two combustors to overall emission levels has been discussed in detail and highlights how the latest EV combustor upgrade has improved the overall engine emission performance. Also, it has been explained how the proven operational flexibility of these engines are inherently linked to the complementary combustion kinetics in the two combustion systems.

The achievement of ultralow emission levels while keeping maximal operational flexibility (both in load variation and fuel composition changes) and reliability will remain the key driver for ALSTOM's combustor development. The sequential combustion system is suited very well for this challenge, both today and for the future.

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